

CS101 - Introduction to Computer Science

Midterm Exam

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Section: 002

Instructions: Answer all questions. Show your work where applicable.

Q1. (10 pts) Explain the difference between a stack and a queue. Give a real-world example of each.

Stack stores data. Queue also stores data.

They are both data structures.

Q2. (10 pts) What is the time complexity of binary search? Explain why.

Binary search is fast. Maybe $O(n)$ or $O(\log n)$ not sure.

Q3. (10 pts) Define recursion and write a recursive function to calculate factorial of n .

Recursion is a loop that repeats.

```
def factorial(n):  
    return n * n
```
